

Repository Bug Collection

Source
Repositories

Bug Screening
& Curation

Bug Evidence Integration & Validation

Empirical Analysis

GitHub

 llama.cpp
 vLLM
 DeepSpeed
 MLC-LLM
 TensorRT-LLM
 TGI

Candidate PR Collection

PR Identifiers & URL

Title, Description
& Labels

Creation Time
& Merge State

Manual Bug-fix Review

Repository Evidence Linking

Merged Bug-fix Eligibility

Eligible Bug Labeling
merged and manually verified
bug-fixing PRs

 **Symptom Assignment**

 **Architectural Layer Assignment**

 **Repair Operation Assignment**

Validated Bug Dataset
  
Validated Bug Artifacts

Failure Localization (RQ1)

Symptom & Architectural Layer Distributions

Symptom-Architectural Layer Lift Associations

Repair Structure (RQ2)

Repair Operations × Architectural Layers

Code Concentration & Fix Scope

Repair Evolution (RQ3)

File-Breadth Evolution

Architectural Composition Evolution